SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – APPLICATION BASED QUESTIONS

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO2	**Assessment:**	Application Based Questions, In-class
Weightage:	_____ 40% of CIA	**Duration / Marks:**	60 mins
CO2 Statement:	Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)		
Student Name:	K Mahesh Sai	**Reg. No.:**	192412674
Section / Batch:	3rd Year	**Date:**	20-10-2026

Code of Conduct

I, K Mahesh Sai, Reg. No. 192412674, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K Mahesh Sai

Instructions

- Read each question carefully before answering (in-Class Assessment)
- Each question is a real-world application scenario. Identify the AI technique that fits, apply it step by step, and justify every decision.
- Questions are drawn from Informed Search (Greedy, A*, heuristics), Local Search and Online Search, Constraint Satisfaction Problems, and Adversarial Search (Minimax, Alpha-Beta, evaluation functions).
- Show intermediate tables (open lists, domains, α/β values) and state the final answer clearly.
- Marks will be awarded using the Application Based Questions rubric: understanding of problem context, application of concepts, problem-solving approach, accuracy of solution and clarity.
- Upload your work in GCR and Github.

Section / Topic	Focus	Question Nos.
1. Informed Search Strategies	Greedy best-first search, A* search, heuristic functions (admissibility, consistency)	**Q1**
2. Local Search, Online Search & Constraint Satisfaction	Local search and optimization problems, online search agent, CSP (MRV, degree heuristic, forward checking)	**Q2, Q3**
3. Game Playing: Adversarial Search	Games, minimax algorithm, alpha-beta pruning, evaluation functions	**Q4**

Rubrics for Application Based Questions

Criteria	Weightage	Level 4 – Exemplary (100% of marks)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Understanding of Problem Context	**20%**	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	**30%**	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	**20%**	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	**20%**	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	**10%**	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Q. No. / Criteria	Understanding of Problem Context (2)	Application of Concepts (3)	Problem-Solving Approach (2)	Accuracy of Solution (2)	Clarity (1)	Sub. Total (10)	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Annexure A – Application Based Questions

Question 1: Campus Delivery Robot – Greedy Best-First vs A*

Topic: Informed search strategies, heuristic functions | 15 minutes

A campus delivery robot must carry a parcel from the Store (S) to the Hostel (G). The one-way road network and the robot's heuristic estimate h(n) of the remaining travel cost to G (in minutes) are given below.

From	To	Cost
S	A	1
S	B	4
A	C	2
A	D	5
B	G	5
C	D	1
D	G	2

Node	S	A	B	C	D	G
h(n)	6	5	3	3	2	0

Assume ties are broken alphabetically and that a node is re-opened if a cheaper path to it is found.

(a) Apply Greedy Best-First Search from S to G. Write the order in which nodes are expanded, the path returned and its total cost.

(b) Apply A* search. Show the open list with g(n), h(n) and f(n) after each expansion, the order of expansion, and the final path and cost.

(c) Check whether h(n) is admissible and consistent by verifying the required condition for every node and every road.

(d) Compare the outputs of (a) and (b). Explain why Greedy Best-First Search returned a different result and which algorithm you would deploy for the robot, with justification.

Question 2: Wi-Fi Access Point Placement – Local Search and Online Search

Topic: Local search algorithms and optimization problems, online search agent | 15 minutes

A technician wants to place a Wi-Fi access point at the best position along a corridor with 10 candidate positions x = 0 to 9. The measured signal quality f(x) (higher is better) is:

x	0	1	2	3	4	5	6	7	8	9
f(x)	2	5	9	6	4	7	12	10	8	3

A move shifts the access point to an adjacent position ($x - 1$ or $x + 1$). Steepest-ascent hill climbing moves to the better neighbour and stops when no neighbour is strictly better.

(a) Run steepest-ascent hill climbing from the start positions x = 0, x = 4 and x = 9. Show each step and the final position and signal quality.

(b) Identify all local maxima and the global maximum. If the start position is chosen uniformly at random from the 10 positions, find the probability that one run reaches the global maximum, the probability that at least one of 3 random-restart runs does, and the expected number of runs needed.

(c) A simulated annealing search is at x = 2 and proposes the move to x = 3. Compute the acceptance probability $e^{(\Delta/T)}$ at T = 2 and at T = 1, and explain what this does to the chance of escaping the local maximum as T falls.

(d) The same search is later done by a robot that cannot see the whole corridor, can only measure f(x) where it stands, and cannot undo exploration for free. Explain why this is an online search problem, and state what the agent must remember to avoid cycling.

Question 3: Exam Timetable Scheduling – Constraint Satisfaction Problem

Topic: Constraint satisfaction problem (MRV, degree heuristic, forward checking) | 15 minutes

Five exams A, B, C, D and E must each be given one slot from {1, 2, 3}. Two exams that share students must not be in the same slot. The pairs sharing students are: A–B, A–C, A–D, B–C, B–D, C–E, D–E. Because of invigilator availability, exam A can only be in slots 1 or 2, and exam E only in slots 1 or 2.

Exam	Initial domain
A	{1, 2}
B	{1, 2, 3}
C	{1, 2, 3}
D	{1, 2, 3}
E	{1, 2}

Rules for the search: choose the variable using Minimum Remaining Values (MRV), break ties with the degree heuristic, then alphabetically; try values in ascending order; use forward checking after every assignment.

(a) Draw the constraint graph (list its edges) and compute the degree of every exam.

(b) Select the first exam to assign using MRV and the degree heuristic, justify the choice, assign its first value, and show the domains of all exams after forward checking.

(c) Continue the search with the same rules, showing the chosen exam, its value and the updated domains at each step, until a complete assignment is found. Verify it against every constraint.

(d) Interpret: explain how MRV, the degree heuristic and forward checking reduced the search effort here, and what forward checking would signal if a domain became empty.

Question 4: Board-Game AI – Minimax, Alpha-Beta Pruning and Evaluation Function

Topic: Game playing, adversarial search, evaluation functions | 15 minutes

A game-playing agent (MAX) searches two plies ahead: MAX chooses one of three moves (B, C, D), then MIN replies with one of three moves. Positions at depth 2 are scored by the evaluation function

$$\mathbf{Eval = 2 \times M + 1 \times Mo}$$

where M is the material advantage of MAX and Mo is the mobility advantage of MAX. The leaf features (M, Mo) are given for the leaves in left-to-right order:

MIN node	Leaf 1 (M, Mo)	Leaf 2 (M, Mo)	Leaf 3 (M, Mo)
B	(1, 1)	(5, 2)	(3, 2)
C	(1, 0)	(1, 2)	(2, 2)
D	(6, 2)	(2, 1)	(0, 2)

(a) Compute the Eval value of all nine leaves.

(b) Run plain minimax. Give the backed-up value of B, C and D, the value at the root and the move MAX should play.

(c) Run alpha-beta pruning on the same tree (left to right). Show α and β at each node, mark every pruned branch, and state the root value and the number of leaves evaluated.

(d) If the leaves under C were ordered (6, 4, 2) instead, how many leaves would alpha-beta evaluate? Explain the effect of move ordering and why pruning never changes the minimax value.

Optimal path: S → A → C → D → G. Total cost = 1 + 2 + 1 + 2 = 6 minutes.

(c) Admissibility and consistency

The shortest remaining costs to G are: h*(S)=6, h*(A)=5, h*(B)=5, h*(C)=3, h*(D)=2, h*(G)=0. The supplied h values are 6, 5, 3, 3, 2, 0 respectively.

Admissibility: h(n) ≤ h*(n) for every node. This holds: B has h(B)=3 ≤ 5; all other heuristic values equal their true shortest remaining costs. Therefore, h is admissible.

Consistency requires h(u) ≤ c(u,v)+h(v) for every directed road:

Road	Check	Result
S→A	6 ≤ 1+5 = 6	Pass
S→B	6 ≤ 4+3 = 7	Pass
A→C	5 ≤ 2+3 = 5	Pass
A→D	5 ≤ 5+2 = 7	Pass
B→G	3 ≤ 5+0 = 5	Pass
C→D	3 ≤ 1+2 = 3	Pass
D→G	2 ≤ 2+0 = 2	Pass

All roads satisfy the consistency condition; h is consistent.

(d) Comparison and deployment

Greedy Best-First Search returns S→B→G at cost 9 because it prefers the node with the smallest h value, G, after expanding B. It ignores the accumulated travel cost g(n), so it misses the cheaper route through A, C and D. A* considers both g and h and returns the 6-minute route. I would deploy A* because the heuristic is admissible and consistent, and A* finds the least-cost route under these conditions.

Question 2: Wi-Fi Access Point Placement – Local Search and Online Search

(a) Steepest-ascent hill climbing

Start	Sequence of positions (x : f(x))	Final position	Final quality
x=0	0:2 → 1:5 → 2:9	2	9
x=4	4:4 → 5:7 → 6:12	6	12
x=9	9:3 → 8:8 → 7:10 → 6:12	6	12

At each step, the move is to the adjacent neighbour with the higher signal quality. Search stops when neither neighbour is strictly better.

(b) Local maxima, global maximum and restart probabilities

Local maxima: x=2 with f(2)=9 and x=6 with f(6)=12. The global maximum is x=6, with quality 12.

Starts that reach x=6 are x=3,4,5,6,7,8,9: 7 of the 10 equally likely starts. Thus P(success in one run)=7/10=0.7.

For 3 independent random-restart runs, P(at least one success)=1−(1−0.7)^3=1−0.3^3=0.973, or 97.3%.

The number of independent runs until the first success is geometric with p=0.7, so the expected number is 1/p=10/7≈1.43 runs.

(c) Simulated annealing

At x=2, f(2)=9; the proposed move to x=3 has f(3)=6. Therefore Δ=f(new)−f(current)=6−9=−3.

Temperature T	Acceptance probability e^(Δ/T)	Approximate value
2	e^(−3/2)	0.2231 (22.31%)

1	e^(−3)	0.0498 (4.98%)

As temperature falls, the probability of accepting this downhill move decreases. A higher temperature makes it more likely to accept worse moves and escape a local maximum; a lower temperature makes the search more selective.

(d) Why this is online search

It is an online search problem because the robot cannot access the complete corridor map and learns signal quality only by measuring its current position while acting. Its actions affect what it can observe next, and exploration cannot be undone for free.

To avoid cycling, it should remember visited positions and relevant measured values, plus the action/state history or a compact map of explored positions. It can then avoid repeatedly moving between the same states.

Question 3: Exam Timetable Scheduling – Constraint Satisfaction Problem

(a) Constraint graph and degrees

Vertices: A, B, C, D, E. Edges (exams sharing students): A–B, A–C, A–D, B–C, B–D, C–E, D–E.

Exam	Neighbours	Degree
A	B, C, D	3
B	A, C, D	3
C	A, B, E	3
D	A, B, E	3
E	C, D	2

(b) First assignment using MRV and degree heuristic

Initial domain sizes: A=2, B=3, C=3, D=3, E=2. MRV ties A and E, both with two values. The degree heuristic breaks the tie: A has degree 3 while E has degree 2, so choose A. Try the lowest value, A=1.

Forward checking removes slot 1 from A's neighbours B, C and D. E is not adjacent to A.

Exam	Domain after A=1
A	{1} (assigned)
B	{2, 3}
C	{2, 3}
D	{2, 3}
E	{1, 2}

(c) Continue the search

Step	Choice and assignment	Domains after forward checking
1	A=1	A={1}; B={2,3}; C={2,3}; D={2,3}; E={1,2}
2	B=2	A={1}; B={2}; C={3}; D={3}; E={1,2}
3	C=3	A={1}; B={2}; C={3}; D={3}; E={1,2}
4	D=3	A={1}; B={2}; C={3}; D={3}; E={1,2}
5	E=1	A={1}; B={2}; C={3}; D={3}; E={1}

Why B is next: after A=1, B, C, D and E all have two remaining values, but B, C and D have degree 3; alphabetical tie-breaking selects B. Assigning B=2 removes 2 from C and D, leaving each with {3}. C and D are not connected by an edge, so both may use slot 3. Finally E can use slot 1 (or 2); the ascending-value rule chooses 1.

Final assignment: A=1, B=2, C=3, D=3, E=1.

Constraint verification: A differs from B, C and D; B differs from C and D; C differs from E; D differs from E. All listed edges have different slots at their endpoints, and A and E both use allowed slots 1 or 2. Therefore the assignment is valid.

(d) Interpretation

MRV chooses the exam with the fewest available slots, exposing constrained variables early. The degree heuristic prioritizes the variable that constrains the largest number of unassigned neighbours when MRV ties. Forward checking immediately removes an assigned value from neighbouring domains, preventing incompatible choices from persisting. If any domain becomes empty, the current partial assignment cannot be extended; the search must backtrack.

Question 4: Board-Game AI – Minimax, Alpha-Beta Pruning and Evaluation Function

(a) Evaluate all nine leaves

Eval = 2 × M + Mo.

MIN node	Leaf features (M, Mo)	Calculation	Eval
B	(1,1)	2×1+1	3
B	(5,2)	2×5+2	12
B	(3,2)	2×3+2	8
C	(1,0)	2×1+0	2
C	(1,2)	2×1+2	4
C	(2,2)	2×2+2	6
D	(6,2)	2×6+2	14
D	(2,1)	2×2+1	5
D	(0,2)	2×0+2	2

(b) Plain minimax

MIN chooses the smallest leaf value under each of its moves: B=min(3,12,8)=3; C=min(2,4,6)=2; D=min(14,5,2)=2.

MAX chooses the largest backed-up value: root=max(3,2,2)=3. MAX should play move B.

(c) Alpha-beta pruning, left to right

Node / event	Leaf value	α	β	Action
Root MAX starts	—	−∞	+∞	Visit B
B MIN, leaf 1	3	−∞	3	β becomes 3
B MIN, leaf 2	12	−∞	3	β remains 3
B MIN, leaf 3	8	−∞	3	B returns 3; root α becomes 3
C MIN, leaf 1	2	3	2	β=2 ≤ α=3; prune C leaves 2 and 3

D MIN, leaf 1	14	3	14	Continue
D MIN, leaf 2	5	3	5	Continue
D MIN, leaf 3	2	3	2	D returns 2

The two branches pruned are C's second and third leaves (values 4 and 6). The root value remains 3, and MAX chooses B. Evaluated leaves: 3 under B + 1 under C + 3 under D = 7 leaves.

(d) Reordering C's leaves to (6, 4, 2)

With C's leaves ordered 6, 4, 2, the MIN node first sees 6 (β=6), then 4 (β=4); neither value is ≤ the root α=3, so no pruning occurs before the third leaf. The third value 2 makes β=2, but there are no leaves remaining under C to prune. Thus all 3 leaves under C are evaluated; B and D also evaluate 3 each, for 9 total leaves.

Move ordering affects efficiency: a low value found early at a MIN node can trigger pruning when it is no greater than MAX's current α. Pruning never changes the minimax value because only branches that cannot improve the final decision are skipped; their evaluation cannot change the root's selected value.

Final Answer Summary

- Q1: Greedy path S→B→G costs 9; A* path S→A→C→D→G costs 6. Heuristic is admissible and consistent.
- Q2: Local maxima x=2 and x=6; global maximum x=6. One-run success probability 0.7; three-restart success probability 0.973; expected runs 10/7.
- Q3: Valid timetable: A=1, B=2, C=3, D=3, E=1.
- Q4: Minimax root value 3, best move B; alpha-beta evaluates 7 leaves in the given order and 9 when C's leaves are ordered 6,4,2.